

Algorithm Analysis And Data Structures

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COURSE CONTENT

- *Introduction of Algorithm Analysis; Mathematical Functions;*
- *Running Time Calculations of C Programs (examples);*
- *Algorithms Analysis Functions (Big Oh, Little Oh, Theta, Omega);*
- *Divide & Conquer Algorithm; Euclid's Algorithm; Binary Search (C codes and examples);*
- *Sorting Algorithms (C codes and examples);*
- *Introduction of Data Structures; Lists (C codes and examples);*
- *Stacks and Queues (C codes and examples);*
- *Data Structures; Trees ; Binary Trees (C codes and examples);*
- *Search Trees (C codes and examples);*
- *AVL Trees (examples);*
- *B & B+ Trees (its algorithms and examples);*

RESOURCES

- *Data Structures and Algorithm Analysis in C*, M. Allen Weiser, ISBN: 0-201-49840-5, Pearson Publications;
- *“Algorithms, Data Structures and Problem Solving with C++”*, Mark Allen Weiss, ISBN: 0-201-36122-1;
- *“Data Structures using C and C++”*, Yedidyah Langsam, M. J. Augenstein, A. M. Tenenbaum;
- *“Algorithms & Complexity”*, Wilf Herberts, Prentice Hall Pub.
- *“Algorithms”*, Cormen L. Risesr, Mc Grawtwill
- *Bilg. Prog. veYaz. Müh. Veri Yapıları ve Algoritmalar”*,
- Rifat Çölkesen, Papatya Yayıncılık, 2002;

RESOURCES

The course detail is given in

www.mtm.yildiz.edu.tr...

WEEKLY COURSE PLAN

Week	Subject
1	<i>Introduction of Algorithm Analysis; Mathematical Functions</i>
2	<i>Algorithms Analysis Functions (Big Oh, Little Oh , Theta , Omega)</i>
3	<i>Running Time Calculations of C Programs (examples)</i>
4	<i>Divide & Conquer Algorithm; Euclid's Algorithm; Binary Search; Estimation of the Given Algorithm's Program Times</i>
5	<i>Sorting Algorithms, C codes and examples</i>
6	<i>Introduction of Data Structures; Lists</i>
7	<i>Lists (C codes and examples)</i>
8	<i>Stacks and Queues (C codes and examples)</i>

WEEKLY COURSE PLAN

Week	Subject
9	<i>Data Structures; Trees ; Binary Trees (C codes and examples)</i>
10	<i>EXAM-II</i>
11	<i>Search Trees; AVL Trees;</i>
12	<i>B Trees and examples</i>
13	<i>EXAM-II</i>
14	<i>B+ Trees and examples</i>
15	<i>Make Up EXAM</i>

EVALUATION

	Number	Effective Proportion %
Midterm Exams	<i>1</i>	<i>60</i>
Quiz		
Homework		
Term Projects		
Term Papers		
Laboratory		
Other		
Final Exam	<i>1</i>	<i>40</i>

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Questions and Answers

Thank you...

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